DDFGFormer: Dual-Decoupled Frequency-Guided Transformer for Hyperspectral Image Classification

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***Abstract*—Hyperspectral image (HSI) classification requires jointly capturing subtle spectral differences and complex spatial structures. Recent studies have introduced frequency-domain modeling to complement spatial–spectral representation learning. However, existing frequency-based methods still face some limitations. First, most methods mainly focus on amplitude or energy distributions while insufficiently exploiting phase information for structural localization and edge representation. In addition, raw phase information is sensitive to noise, smooth regions, and mixed pixels, which may introduce unreliable interference. Furthermore, spatial–frequency interaction is often implemented through simple posterior fusion, failing to use frequency-domain priors to actively guide spatial feature extraction. To address these limitations, we propose a dual-decoupled frequency-guided transformer, termed DDFGFormer, for HSI classification. In particular, we design a frequency-guided transformer block (FGTB) composed of dual-decoupled frequency attention (DDFA) and a frequency-enhanced feed-forward network (FEFFN). DDFA decouples the Fourier representation into amplitude and phase, and further decomposes the amplitude into high- and low-frequency components, thereby explicitly capturing amplitude- and phase-aware cues. To suppress unreliable phase responses, an amplitude-prior guided phase (APGP) module contained in DDFA is proposed to adaptively modulate phase representations and transform raw phase into stable structural priors. Furthermore, FEFFN injects frequency-domain information into feature propagation, enabling frequency priors to actively recalibrate spatial features rather than serving only as posterior fusion cues. Extensive experiments on four HSI datasets show that DDFGFormer outperforms nine state-of-the-art methods in overall accuracy (OA), average accuracy (AA), kappa, and boundary $F1$-score (BF1), with OA gains of 1.65%, 1.63%, 1.25%, and 1.14% over the second-best method on Indian Pines (IPs), Pavia University (PU), Salinas (SAs), and WHU-Hi-LongKou (LK), respectively.**

***Index Terms*—Frequency-domain modeling, hyperspectral image (HSI) classification, phase information, transformer.**

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I. Introduction

HYPERSPECTRAL images (HSIs) record fine-grained spectral responses of ground objects over hundreds of contiguous bands. Compared with RGB images, HSIs contain rich spectral signatures as well as spatial structural information, thereby providing strong material discrimination capability. Nevertheless, accurate HSI classification remains challenging due to high-dimensional spectral redundancy, subtle interclass spectral discrepancies, intraclass spectral variability, and boundary mixing in complex scenes [1].

Recently, deep learning-based HSI classification methods have drawn increasing attention [2], [3]. Convolutional neural network (CNN)-based methods are effective in extracting local spatial–spectral patterns and have achieved remarkable progress in HSI classification [4]. However, their local receptive fields limit their ability to model long-range dependencies and large-scale contextual relationships [5]. Transformers alleviate this limitation through self-attention and have demonstrated strong global contextual modeling capability in HSI classification [6]. Despite their global modeling ability, transformer-based methods may weaken local details and boundary structures during global feature aggregation, which can degrade the discrimination of fine-grained categories and structurally complex regions [7]. To balance local detail extraction and global perception, many studies have adopted dual-branch, hybrid, or fusion architectures that combine CNNs, transformers, graph models, or other global modeling modules [8], [9], [10]. Although these methods improve spatial–spectral representation from different backbone designs, they still mainly operate in the original spatial domain and do not explicitly exploit frequency-domain structural cues that are closely related to edges, textures, and local variations.

Frequency-domain analysis provides a complementary perspective for addressing this limitation, since the frequency domain represents variation patterns, contour structures, and detail responses that are not explicitly captured in the spatial domain [11], [12], [13]. Existing frequency-aware HSI methods can be organized along three research directions. The first direction concentrates on amplitude-based frequency decomposition and band modeling [14]. By transforming spatial–spectral features into the Fourier domain and separating high-frequency components from low-frequency ones, these methods capture fine-grained local details and global

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smooth variations through distinct processing pathways [15]. Subsequent efforts within this direction have introduced multi-band partitioning, frequency-aware prompts, and cross-scale frequency aggregation to enhance discriminative power [16]. Despite their effectiveness, this line of work operates almost exclusively on amplitude spectra, and phase information, which encodes structural localization, spatial arrangement, and edge positions, is not explicitly exploited. The second direction moves beyond amplitude-centric modeling by investigating the role of phase [17]. Phase reconstruction has been shown to reveal clear structural contours; preserving phase while perturbing amplitude helps maintain geometric consistency in cross-scene classification, and phase-aware filtering as well as amplitude-phase interaction mechanisms provide complementary discriminative cues beyond amplitude-only representations [18]. These findings confirm that phase carries structurally meaningful information. However, existing phase-aware methods treat phase as an intrinsically usable signal without addressing a fundamental difficulty. Raw phase is highly sensitive to noise, homogeneous regions, and mixed pixels, and directly incorporating it can introduce unstable interference rather than consistent structural enhancement. The third direction concerns how frequency-domain features are combined with spatial features [19]. Regardless of whether the frequency modeling is amplitude-only or phase-aware, existing methods predominantly adopt posterior fusion strategies in which frequency-domain and spatial features are first extracted independently and then merged through concatenation, branch weighting, or cross-domain attention [20]. Under this paradigm, frequency-domain information serves as an auxiliary supplement rather than an explicit guidance signal that actively directs spatial feature learning.

Motivated by the above observations, we propose a dual-decoupled frequency-guided transformer, termed DDFGFormer, for HSI classification. The core idea of DDFGFormer is to explicitly model frequency-domain representations at two levels and use reliable frequency cues to guide spatial feature refinement. In particular, the Fourier representation is first decoupled into amplitude and phase components. The amplitude component is further decomposed into high- and low-frequency responses, allowing the model to separately capture local details and global contours. Since high-frequency amplitude responses usually indicate structurally salient regions such as edges, textures, and abrupt spectral–spatial variations, they can serve as reliability priors for phase modeling. Based on this observation, we design an amplitude-prior guided phase (APGP) module to adaptively modulate phase representations using high-frequency structural responses. This mechanism enhances phase encoding in structurally clear regions while suppressing unreliable phase interference in noise-dominated regions. Furthermore, we design a frequency-enhanced feed-forward network (FEFFN) to continuously inject discriminative frequency-domain information into feature propagation, thereby improving spatial feature representation and classification robustness in complex scenes.

The main contributions of this article are summarized as follows.

1) We propose a dual-decoupled frequency-domain modeling strategy for HSI classification. By decoupling Fourier features into amplitude and phase and further decomposing the amplitude component into high- and low-frequency responses, the proposed strategy enables separate modeling of local details, global contours, and phase-aware structural localization cues.
2) We propose an APGP module, which uses high-frequency amplitude responses as reliability priors to adaptively modulate phase representations. This design improves the utilization of phase-aware structural information while alleviating the instability caused by direct phase modeling.
3) We design an FEFFN, which introduces discriminative frequency-domain information during feature propagation. Together with the proposed frequency-guided attention mechanism, FEFFN enhances spatial feature refinement and improves classification performance for fine-grained categories and structurally complex regions.

The remainder of this article is organized as follows. Section II systematically reviews related work; Section III presents the proposed DDFGFormer in detail; Section IV demonstrates experimental results; and Section V concludes this article and discusses directions for future research.

II. Related Work

A. Spatial–Spectral-Aware Methods for HSI Classification

The core of HSI classification lies in the joint exploitation of spatial and spectral information for fine-grained land-cover discrimination [21]. Existing spatial–spectral-aware methods can be roughly divided into CNN-based, transformer-based, and hybrid modeling frameworks [22], [23]. CNN-based methods exploit local connectivity to capture spatial textures within neighboring regions. For instance, SGDCN [24] introduces superpixel segmentation to preserve object-level spatial homogeneity and leverages deformable convolution to adaptively adjust receptive fields. Nevertheless, CNNs are inherently constrained by local receptive fields, which limits their ability to model long-range contextual relationships and cross-region semantic dependencies in complex HSI scenes.

To overcome this limitation, transformer-based methods have been introduced into HSI classification because self-attention can establish global interactions among tokens. SSFTT [25] utilizes shallow 3-D and 2-D convolutions to smooth the transition from low-level local spatial–spectral features to high-level global semantic representations. Other transformer variants further improve this paradigm through hierarchical embedding or spatial–spectral self-attention [26], [27]. Although these methods enhance global dependence modeling, the global aggregation process of self-attention inherently acts as a low-pass filter, which tends to smooth out high-frequency details and weaken local boundary structures critical for fine-grained categories [28].

To balance local detail preservation and global context modeling, hybrid CNN-transformer architectures have been widely explored. For example, CTMixer [29] routes inputs into concurrent convolutional and attention paths to achieve multiscale fusion, whereas SWFormer [30] introduces stochastic window-based modeling to regularize hierarchical representations. Beyond architectural fusion, another line of work addresses the related challenges of cross-domain shifts and

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sample scarcity. For instance, OFD-Net [31] utilizes orthogonal subspace decomposition to project spatial–spectral features into mutually exclusive domain-shared and -specific streams, successfully suppressing domain-specific noise and ensuring structural consistency under target domain shifts. Similarly, SAEPL [32] departs from conventional mean-based prototype learning by introducing orthogonally initialized trainable semantic anchors and contrastive objectives to explicitly align class prototypes across heterogeneous source and target distributions. Overall, these methods have significantly advanced HSI classification within the spatial–spectral domain. However, they still mainly operate in the original spatial–spectral domain and lack explicit exploitation of frequency-domain structural cues, such as edges, textures, and local variations. This motivates the introduction of frequency-domain analysis to complement spatial–spectral representation learning.

B. Frequency-Aware Methods for HSI Classification

Frequency-domain modeling provides a complementary perspective for enhancing boundary-sensitive and detail-aware HSI representations [33]. Existing frequency-aware methods can be organized along three research threads, each focusing on a distinct aspect of frequency-domain representation learning.

1) Amplitude-Based Modeling and Band-Level Enhancement: The first research thread focuses on amplitude- or energy-distribution modeling, constructing frequency representations based on the functional differences between frequency bands [34], [35]. A pioneer in this line is the dual frequency transformer network proposed by Qiao and Huang [36], which develops a dual-branch feature extraction block where a multihead neighborhood attention module and a global filter module are explicitly tailored to capture high-frequency details and low-frequency smooth variations, respectively. Subsequent efforts, such as SFDCT-former [37], extend this by dividing the frequency domain into nonoverlapping bands and assigning distinct attention weights to accommodate individual band characteristics. To prevent cross-scale informational loss, FS-CGNet [38] employs fast Fourier transform (FFT) techniques within a spectral-channel fusion framework to aggregate cross-scale frequency features. Furthermore, LFAH-Net [39] introduces local frequency-adaptive gating to dynamically reweight amplitude coefficient energy, while DFAST [40] deploys a dual-branch frequency-attention mechanism to explicitly route amplitude cues into band-selection networks. These studies demonstrate the usefulness of amplitude-related frequency information for HSI classification. However, they mainly focus on frequency energy distributions or band-level enhancement, while the phase component, which is closely related to structural localization and edge arrangement, remains entirely unexploited.

2) Wavelet-Based Multiscale Frequency Analysis: The second research thread focuses on wavelet-based multiscale frequency analysis [41]. Unlike global Fourier representations, wavelet transforms decompose features into localized multiscale subbands, making them highly suitable for preserving local textures and boundary details [42]. For instance, WaveFormer [43] incorporates discrete wavelet transforms into reversible downsampling paths within transformer encoders, effectively mitigating pooling-induced information loss and enhancing structural fidelity. To capture even finer topologies, MWGN-SSA [44] develops a multiscale learnable wavelet network (MLWN) that adaptively decomposes HSIs into discrete frequency subbands, successfully retaining fine-grained high-frequency textures for spectrally subtle or small targets. Recently, this multiscale paradigm has been seamlessly integrated into state-space backbone architectures to combine efficient long-range scanning with localized frequency perception. A prominent example is SFMamba [45], which designs a wavelet Mamba block (WMB) that utilizes wavelet transform to decompose input features into one low-frequency component and three high-frequency components; while the low-frequency structure is processed via standard convolution, the texturally rich high-frequency components are directly assigned to a Mamba block for global sequential modeling. Similarly, WD-SSMamba [46] incorporates both 1-D and 2-D wavelet decompositions into dedicated spectral and spatial feature extraction paths to regularize Mamba state transitions. Nevertheless, while these methods effectively enhance multiscale structural representation, their formulations remain fundamentally amplitude-bounded. They treat frequency information strictly as subband-level coefficients or energy distributions, leaving the crucial phase reliability problem completely unaddressed in this multiscale regime.

3) Phase-Related and Amplitude-Phase-Aware Modeling: The third research thread concerns phase-related or amplitude-phase-aware frequency modeling. Amplitude and phase represent two fundamental, complementary components of Fourier-domain signatures, where amplitude reflects variation intensity and phase encodes geometric arrangement and contour positions. Recent studies have initiated explorations into phase-related properties. In particular, Tu et al. [47] introduce a quaternion frequency-domain reconstruction method for hyperspectral anomaly detection, demonstrating through phase-only reconstruction that while amplitude spectra characterize dominant backgrounds, phase spectra are uniquely critical for perceiving spatial saliency. To mitigate distribution shifts in cross-scene HSI classification, Lin et al. [48] propose a domain generalization approach that strategically perturbs amplitude spectra while preserving phase to enforce structural and geometric consistency across different target domains. In addition, phase-induced Gabor filtering has been explored to provide complementary discriminative cues under limited training samples [49]. These studies verify that phase carries valuable structural information that amplitude alone cannot provide. Critically, however, raw phase is intrinsically sensitive to noise, homogeneous regions, and mixed pixels. Existing methods usually preserve, reconstruct, or encode phase information directly, but they rarely estimate when and where phase responses are reliable, leaving the fragile phase signals highly vulnerable to unstable structural interference.

III. Proposed Method

A. Overall Architecture

DDFGFormer explicitly incorporates frequency-domain analysis into a transformer-style framework. As illustrated in Fig. 1, the proposed dual-decoupled frequency-guided transformer (DDFGFormer) consists of three major parts: a shallow feature extraction (SFE) module, a three-stage hierarchical spatial–frequency backbone, and a lightweight

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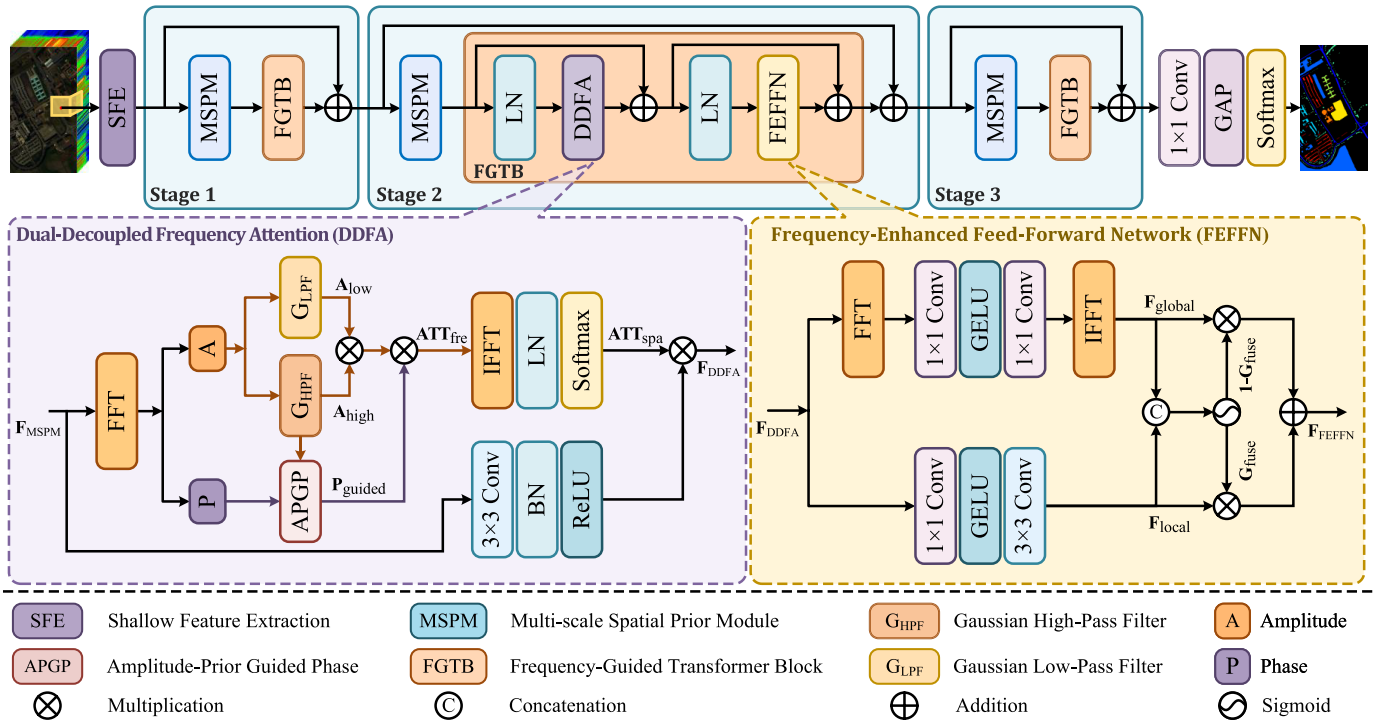


Fig. 1. Overview of the proposed DDFGFormer. DDFGFormer adopts a multistage hierarchical architecture to jointly model spatial and frequency-domain information. It comprises an SFE followed by three stages, each integrating MSPM and FGTB for multiscale spatial context modeling and frequency-aware representation learning. Within FGTB, the DDFA module decouples amplitude and phase for separate modeling, while the FEFFN further enhances features through spatial–frequency fusion. Finally, a classification head generates the prediction.

prediction head. In particular, given an original HSI, we extract spatial neighborhoods centered on each target pixel to form input patches $\mathbf{X} \in \mathbb{R}^{p\times p\times c}$, where $p \times p$ denotes the spatial size of each patch and c denotes the number of spectral bands. Each input patch is first projected into a compact latent space by SFE, which suppresses redundant spectral responses and stabilizes local spatial–spectral patterns. The resulting shallow representation is then fed into three consecutive stages. In each stage, a prior module, multiscale spatial (MSPM), is first used to strengthen local spatial priors, after which a frequency-guided transformer block (FGTB) performs frequency-aware structural refinement. In particular, FGTB introduces a dual-decoupled frequency attention (DDFA) module to decouple Fourier responses into amplitude and phase and further separate the amplitude branch into high- and low-frequency components, thereby enabling the model to jointly exploit local textures, global contours, and phase-aware structural information. Furthermore, an APGP mechanism is introduced to improve the reliability of phase-aware structural cues. In addition, an FEFFN continuously injects discriminative frequency-domain information during feature propagation. Finally, the refined representation is globally aggregated and projected to the label space for prediction.

B. Shallow Feature Extraction

Although hyperspectral patches contain abundant spectral responses, they also suffer from strong redundancy, local variability, and mixed-pixel interference. Directly performing frequency-domain modeling on the raw input may amplify unstable fluctuations and irrelevant responses. Therefore, we first introduce an SFE module to map the raw patch into a compact and discriminative latent space. SFE is implemented by two stacked convolutional layers, each followed by batch normalization and nonlinear activation

$$\mathbf{F}_0 = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times3}\left(\mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times1}\left(\mathbf{X}\right)\right)\right)\right)\right)\right) \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{p\times p\times c}$ is the input patch and $\mathbf{F}_0 \in \mathbb{R}^{p\times p\times d}$ denotes the shallow feature, where $d = 64$ is the embedding dimension. $\mathrm{Conv}_{1\times1}(\cdot)$ and $\mathrm{Conv}_{3\times3}(\cdot)$ denote convolutional operations with 1×1 and 3×3 kernels, respectively. All convolutional layers in this work employ appropriate zero padding to preserve the spatial resolution of feature maps. $\mathrm{BN}(\cdot)$ represents batch normalization, which normalizes intermediate features to accelerate convergence and improve training stability. $\mathrm{ReLU}(\cdot)$ denotes the rectified linear unit activation function. This operation suppresses redundant spectral responses while preserving local spatial–spectral patterns, thereby providing a stable feature basis for subsequent stage-wise refinement.

C. Multiscale Spatial Prior Module

The backbone of DDFGFormer is composed of three successive stages. Each stage contains an MSPM followed by a FGTB. The former is responsible for strengthening local spatial priors, whereas the latter performs frequency-guided structural refinement. Although DDFGFormer emphasizes frequency-guided modeling, local spatial–spectral cues remain essential for HSI classification. In particular, neighboring structures, local textures, and multiscale boundaries provide important evidence for distinguishing classes with highly

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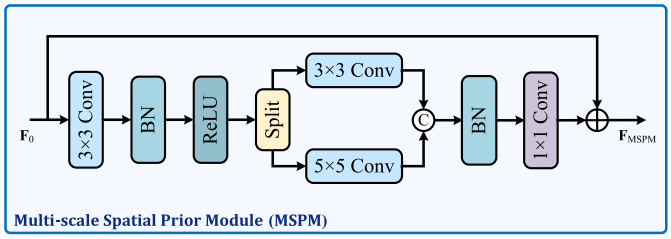


Fig. 2. Architecture of MSPM.

similar spectral signatures. To this end, we introduce MSPM to explicitly extract multiscale spatial features before frequency-domain refinement.

As shown in Fig. 2, given an input feature $\mathbf{F}_0 \in \mathbb{R}^{p\times p\times d}$, MSPM first performs preliminary spatial encoding

$$\mathbf{F}_{\text{mid}} = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{3\times 3}\left(\mathbf{F}_0\right)\right)\right). \tag{2}$$

Then, the encoded feature $\mathbf{F}_{\text{mid}} \in \mathbb{R}^{p\times p\times d}$ is evenly split into two channel groups $\mathbf{F}_1 \in \mathbb{R}^{p\times p\times d/2}$ and $\mathbf{F}_2 \in \mathbb{R}^{p\times p\times d/2}$. Subsequently, $\mathbf{F}_1$ and $\mathbf{F}_2$ are processed by two convolutional branches with different receptive fields and concatenated as follows:

$$\mathbf{F}_{\text{hier}} = \text{Concat}\left(\text{Conv}_{3\times 3}\left(\mathbf{F}_1\right), \text{Conv}_{5\times 5}\left(\mathbf{F}_2\right)\right) \tag{3}$$

where Concat$(\cdot)$ denotes channelwise concatenation. The resulting feature $\mathbf{F}_{\text{hier}} \in \mathbb{R}^{p\times p\times d}$ is further refined by a 1×1 convolution for channelwise feature fusion. The refined feature is then combined with the original input through a residual connection to preserve spatial information and improve optimization stability

$$\mathbf{F}_{\text{MSPM}} = \text{Conv}_{1\times 1}\left(\text{BN}\left(\mathbf{F}_{\text{hier}}\right)\right) + \mathbf{F}_0. \tag{4}$$

Through this hierarchical multiscale design, MSPM captures both local textures and coarse contours, thereby providing a stronger spatial prior for the subsequent frequency-guided refinement.

D. Frequency-Guided Transformer Block

Compared with a classical transformer block, FGTB does not simply rely on self-attention and pointwise transformation in the spatial domain. Instead, it integrates frequency-domain analysis into both the attention stage and the feature transformation stage, enabling the network to jointly exploit local textures, global contours, and structural phase information. As shown in Fig. 1, each FGTB contains two core modules: DDFA and FEFFN.

Given the MSPM output $\mathbf{F}_{\text{MSPM}} \in \mathbb{R}^{p\times p\times d}$ as the input to FGTB, FGTB first applies layer normalization and DDFA, followed by a residual connection

$$\mathbf{F}_{\text{DDFA}} = \text{DDFA}\left(\text{LN}\left(\mathbf{F}_{\text{MSPM}}\right)\right) + \mathbf{F}_{\text{MSPM}} \tag{5}$$

where LN$(\cdot)$ denotes layer normalization. The intermediate feature is further refined by FEFFN

$$\mathbf{F}_{\text{FGTB}} = \text{FEFFN}\left(\text{LN}\left(\mathbf{F}_{\text{DDFA}}\right)\right) + \mathbf{F}_{\text{DDFA}}. \tag{6}$$

In this design, DDFA is responsible for constructing structurally informed attention from frequency-domain representations, whereas FEFFN further propagates such information through a gated refinement pathway. Therefore, FGTB forms a complete frequency-guided representation learning mechanism rather than a simple posterior fusion design.

E. Dual-Decoupled Frequency Attention

1) Amplitude-Phase Decoupling: Different from conventional self-attention that primarily operates on spatial–spectral features, DDFA is designed to explicitly exploit high-frequency, low-frequency, and phase information in the Fourier domain, thereby providing frequency-aware guidance for subsequent feature refinement.

Given an input feature $\mathbf{F}_{\text{MSPM}} \in \mathbb{R}^{p\times p\times d}$, we first perform a channelwise 2-D FFT as follows:

$$\mathbf{F}_{\text{freq}} = \text{FFT}\left(\text{LN}\left(\mathbf{F}_{\text{MSPM}}\right)\right) \tag{7}$$

where FFT$(\cdot)$ denotes the 2-D Fourier transform independently applied to each feature channel. The resulting frequency-domain representation contains both amplitude and phase information. To leverage their complementary properties, DDFA decomposes $\mathbf{F}_{\text{freq}} \in \mathbb{R}^{p\times(\lfloor p/2\rfloor+1)\times d}$ into distinct amplitude and phase components. In particular, while the amplitude reflects the intensity distribution of frequency responses, the phase preserves key spatial structural information, such as contours and edges. These two components are computed as follows:

$$\mathbf{A} = |\mathbf{F}_{\text{freq}}|, \qquad \mathbf{P} = \angle\left(\mathbf{F}_{\text{freq}}\right) \tag{8}$$

where $|\cdot|$ and $\angle(\cdot)$ denote the amplitude and phase operators, respectively.

2) High- and Low-Frequency Decomposition: Although amplitude already reflects the strength and distribution of spectral–spatial responses, it still mixes globally smooth structures with fine local details. Therefore, we further decompose the amplitude branch into low- and high-frequency components, so that coarse contour information and detail-sensitive structural responses can be modeled separately for subsequent guidance and refinement.

In particular, a Gaussian low-pass filter $\mathbf{G}_{\text{LPF}}$ and its complementary Gaussian high-pass filter $\mathbf{G}_{\text{HPF}}$ are employed to decompose the amplitude component. Compared with hard frequency partitioning, Gaussian filtering provides smoother frequency transitions and reduces abrupt changes near the frequency boundary. Using normalized frequency coordinates (u, v), the low-pass filter is defined as

$$\mathbf{G}_{\text{LPF}} = \exp\left(-\frac{u^2+v^2}{2\sigma^2}\right) \tag{9}$$

where σ denotes the normalized bandwidth parameter and controls the transition between low- and high-frequency responses. The high-pass filter is defined as the complementary filter

$$\mathbf{G}_{\text{HPF}} = 1 - \mathbf{G}_{\text{LPF}}. \tag{10}$$

In all experiments, the bandwidth parameter is fixed as $\sigma = 0.20$. The low- and high-frequency amplitude components are then obtained as

$$\mathbf{A}_{\text{low}} = \mathbf{A} \odot \mathbf{G}_{\text{LPF}}, \quad \mathbf{A}_{\text{high}} = \mathbf{A} \odot \mathbf{G}_{\text{HPF}} \tag{11}$$

where "$\odot$" indicates elementwise multiplication. To explicitly encode the interaction between globally smooth contours and

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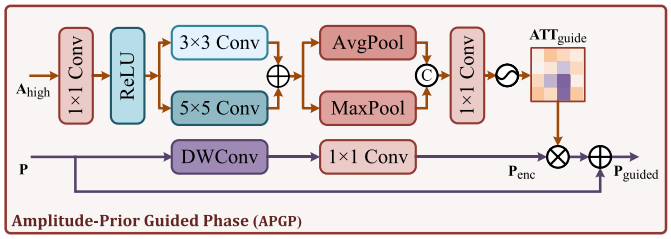


Fig. 3. Architecture of APGP module.

detail-sensitive local structures, we introduce the following multiplicative interaction operation:

$$\mathbf{A}_{\text{int}} = \mathbf{A}_{\text{low}} \odot \mathbf{A}_{\text{high}}. \tag{12}$$

3) Amplitude-Prior Guided Phase: Compared with the amplitude branch, which mainly reflects the strength and distribution of spectral–spatial responses, the phase branch preserves more explicit structural positional information. However, directly using raw phase is often unreliable in smooth, low-SNR, or mixed-pixel regions because of its high sensitivity to local perturbations. On the contrary, the high-frequency amplitude component $\mathbf{A}_{\text{high}}$ provides a more stable indication of structurally salient regions. Therefore, we introduce an APGP mechanism to stabilize phase responses with high-frequency amplitude priors and convert raw phase into a more reliable structural cue.

In particular, as shown in Fig. 3, the high-frequency amplitude component $\mathbf{A}_{\text{high}}$ is first projected into a latent feature space through a 1×1 convolution, followed by nonlinear activation. Then, two parallel convolutional branches with different receptive fields are employed to capture multiscale structural responses

$$\begin{aligned}\mathbf{A}_{\text{mid}} &= \text{ReLU}\left(\text{Conv}_{1\times1}\left(\mathbf{A}_{\text{high}}\right)\right)\\ \mathbf{A}_{\text{ms}} &= \text{Conv}_{3\times3}\left(\mathbf{A}_{\text{mid}}\right) + \text{Conv}_{5\times5}\left(\mathbf{A}_{\text{mid}}\right)\end{aligned} \tag{13}$$

where $\mathbf{A}_{\text{mid}} \in \mathbb{R}^{p\times(\lfloor p/2\rfloor+1)\times d}$ denotes the intermediate amplitude feature projected from the high-frequency component, and $\mathbf{A}_{\text{ms}} \in \mathbb{R}^{p\times(\lfloor p/2\rfloor+1)\times d}$ represents the multiscale amplitude feature obtained by aggregating local structural information with different receptive fields. Then, average pooling and max pooling are jointly adopted to aggregate complementary structural statistics, and the resulting feature is converted into a gating map

$$\begin{aligned}\mathbf{ATT}_{\text{guide}} = \phi\Big(\text{Conv}_{1\times1}\big(\text{Concat}\big(&\text{AvgPool}\left(\mathbf{A}_{\text{ms}}\right)\\ &\times \text{MaxPool}\left(\mathbf{A}_{\text{ms}}\right)\big)\big)\Big)\end{aligned} \tag{14}$$

where $\phi(\cdot)$ denotes the Sigmoid function. In this way, $\mathbf{ATT}_{\text{guide}} \in \mathbb{R}^{p\times(\lfloor p/2\rfloor+1)}$ serves as a high-frequency-guided reliability estimator, highlighting regions where phase responses are more likely to be structurally reliable.

Meanwhile, the raw phase $\mathbf{P}$ is encoded by a phase branch $\mathbf{P}_{\text{enc}} = \text{Conv}_{1\times1}(\text{DWConv}(\mathbf{P}))$, where $\text{DWConv}(\cdot)$ denotes depthwise convolution. The guided phase is then obtained by

$$\mathbf{P}_{\text{guided}} = \mathbf{P}_{\text{enc}} \odot \mathbf{ATT}_{\text{guide}} + \mathbf{P}. \tag{15}$$

In this way, APGP converts unstable raw phase into a robust structure-aware prior. To explicitly incorporate reliable structural positional information into the frequency-domain representation, we use the guided phase response as the phase component for complex-spectrum reconstruction. In particular, the amplitude interaction term $\mathbf{A}_{\text{int}}$ is regarded as the reconstructed amplitude response, while the APGP-enhanced phase $\mathbf{P}_{\text{guided}}$ is constrained to a valid phase range and used to form the complex-valued frequency attention representation

$$\mathbf{ATT}_{\text{fre}} = \mathbf{A}_{\text{int}} \odot \exp\left(j\,\text{Wrap}\left(\mathbf{P}_{\text{guided}}\right)\right) \tag{16}$$

where $\mathbf{ATT}_{\text{fre}}$ denotes the reconstructed frequency-domain attention representation, j is the imaginary unit, and $\text{Wrap}(\cdot)$ maps the phase response into the range of $[-\pi, \pi]$. In this manner, the reconstructed frequency attention preserves the distinct Fourier-domain roles of amplitude and phase, while jointly encoding coarse-to-fine amplitude consistency and phase-aware structural localization cues.

4) Frequency-Aware Attention Reconstruction: To use the reconstructed frequency response for spatial feature refinement, we transform it back to the spatial domain and further normalize it into a frequency-aware attention map

$$\mathbf{ATT}_{\text{spa}} = \text{Softmax}\left(\text{LN}\left(\text{IFFT}\left(\mathbf{ATT}_{\text{fre}}\right)\right)\right) \tag{17}$$

where $\mathbf{ATT}_{\text{spa}} \in \mathbb{R}^{p\times p}$ denotes the generated frequency-aware attention map, and $\text{Softmax}(\cdot)$ normalizes the spatial responses into attention weights to highlight structurally informative regions. This operation converts the reconstructed frequency-domain response into an explicit spatial guidance signal, enabling subsequent feature refinement to be selectively modulated by structurally informative frequency cues.

In parallel, a spatial branch is generated from the input feature $\mathbf{F}_{\text{MSPM}}$ and modulated by the frequency-aware gating map. Furthermore, a residual connection is introduced to make the DDFA preserve the original spatial information

$$\begin{aligned}\mathbf{F}_{\text{DDFA}} = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{3\times3}\left(\text{LN}\left(\mathbf{F}_{\text{MSPM}}\right)\right)\right)\right) \odot \mathbf{ATT}_{\text{spa}}\\ + \mathbf{F}_{\text{MSPM}}.\end{aligned} \tag{18}$$

For clarity, (18) details the concrete implementation of the DDFA module defined in (5) by expanding its internal operations.

In this way, the reconstructed frequency cues are not retained in the frequency domain, but are converted into an explicit structural guidance signal for spatial feature refinement, which is fundamentally different from simple posterior fusion strategies.

F. Frequency-Enhanced Feed-Forward Network

Compared with a standard FFN, FEFFN better balances local texture preservation and global frequency modeling, thereby improving the refinement of fine-grained and structurally complex categories. Given the input feature $\mathbf{F}_{\text{DDFA}}$ from DDFA, a local branch and a global frequency branch are constructed. The local branch focuses on preserving local textures and neighborhood information

$$\mathbf{F}_{\text{local}} = \text{Conv}_{3\times3}\left(\text{GELU}\left(\text{Conv}_{1\times1}\left(\text{LN}\left(\mathbf{F}_{\text{DDFA}}\right)\right)\right)\right) \tag{19}$$

where $\mathbf{F}_{\text{local}}$ denotes the local texture-enhanced feature. The local branch preserves local details, while the global branch is designed to capture complementary global patterns in the frequency domain. In particular, the input feature is transformed by FFT, refined with lightweight pointwise convolutions, and

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TABLE I
DETAILS OF EACH LAND-COVER CLASS OF FOUR DATASETS USED FOR EVALUATION IN THE EXPERIMENTS

No.	Indian Pines			Pavia University			Salinas			WHU-Hi-LongKou		
	Class Name	Train./Val.	Test.	Class Name	Train./Val.	Test.	Class Name	Train./Val.	Test.	Class Name	Train./Val.	Test.
1	Alfalfa	2/2	42	Asphalt	33/33	6565	Brocoli_green_weeds_1	10/10	1989	Corn	173/173	34165
2	Corn-notill	71/71	1286	Meadows	93/93	18463	Brocoli_green_weeds_2	19/19	3688	Cotton	42/42	8290
3	Corn-mintill	41/41	748	Gravel	10/10	2079	Fallow	10/10	1956	Sesame	15/15	3001
4	Corn	12/12	213	Trees	15/15	3034	Fallow_rough_plow	7/7	1380	Broad-leaf soybean	316/316	62580
5	Grass-pasture	24/24	435	Painted Metal Sheets	7/7	1331	Fallow_smooth	13/13	2652	Narrow-leaf soybean	21/21	4109
6	Grass-trees	37/37	656	Bare soil	25/25	4979	Stubble	20/20	3919	Rice	59/59	11736
7	Grass-pasture-mowed	1/1	26	Bitumen	7/7	1316	Celery	18/18	3543	Water	335/335	66386
8	Hay-windrowed	24/24	430	Self-Blocking Bricks	18/18	3646	Grapes_untrained	56/56	11159	Roads and houses	36/36	7052
9	Oats	1/1	18	Shadows	5/5	937	Soil_vinyard_develop	31/31	6141	Mixed weed	26/26	5177
10	Soybean-notill	48/48	876				Corn_senesced_green_weeds	16/16	3246			
11	Soybean-mintill	123/123	2209				Lettuce_romaine_4wk	5/5	1058			
12	Soybean-clean	30/30	533				Lettuce_romaine_5wk	10/10	1907			
13	Wheat	10/10	185				Lettuce_romaine_6wk	5/5	906			
14	Woods	63/63	1139				Lettuce_romaine_7wk	5/5	1060			
15	Buildings-Grass-Trees-Drives	19/19	348				Vinyard_untrained	36/36	7196			
16	Stone-Steel-Towers	5/5	83				Vinyard_vertical_trellis	9/9	1789			

then mapped back to the spatial domain to produce a global frequency-aware representation

$$\mathbf{F}_{\text{global}} = \text{IFFT}\left(\text{Conv}_{1\times1}\left(\text{GELU}\left(\text{Conv}_{1\times1} \times \left(\text{FFT}\left(\text{LN}\left(\mathbf{F}_{\text{DDFA}}\right)\right)\right)\right)\right)\right) \tag{20}$$

where $\mathbf{F}_{\text{global}}$ represents the frequency-enhanced global feature.

Instead of directly concatenating the two branches, FEFFN adopts a gated fusion framework to adaptively balance local and global information

$$\begin{aligned}\mathbf{G}_{\text{fuse}} &= \phi\left(\text{Conv}_{1\times1}\left(\text{Concat}\left(\mathbf{F}_{\text{local}}, \mathbf{F}_{\text{global}}\right)\right)\right)\\ \mathbf{F}_{\text{FEFFN}} &= \mathbf{F}_{\text{local}} \odot \mathbf{G}_{\text{fuse}} + \mathbf{F}_{\text{global}} \odot (1 - \mathbf{G}_{\text{fuse}}).\end{aligned} \tag{21}$$

Through this gated design, FEFFN preserves local texture details while continuously injecting global frequency-domain information into feature propagation.

G. Prediction Head

After the final stage of the backbone, the refined feature map is fed into a classification head for prediction. In particular, the output feature $\mathbf{F}_{\text{out}}$ is first adjusted by a 1×1 convolution and then aggregated by global average pooling. The predicted label $\mathbf{Y}$ is obtained by a linear classifier followed by Softmax:

$$\mathbf{Y} = \text{Softmax}\left(\text{Linear}\left(\text{GAP}\left(\text{Conv}_{1\times1}\left(\mathbf{F}_{\text{out}}\right)\right)\right)\right). \tag{22}$$

Overall, DDFGFormer follows a coherent spatial-to-frequency refinement paradigm. SFE first constructs a stable shallow representation from the raw hyperspectral patch. Then, in each stage of the hierarchical backbone, MSPM strengthens local spatial priors, while FGTB performs frequency-guided refinement. Within FGTB, DDFA explicitly decouples amplitude and phase, further decomposes amplitude into high- and low-frequency components, and stabilizes the phase branch through APGP. FEFFN subsequently propagates such frequency-aware cues through gated refinement. In this way, DDFGFormer can jointly exploit local details, global contours, and phase-aware structural information for robust HSI classification.

IV. EXPERIMENTAL RESULTS

A. Datasets Description

To comprehensively evaluate the performance and generalization capability of the proposed DDFGFormer, extensive experiments are conducted on four widely recognized and publicly accessible HSI benchmarks: Indian Pines (IPs), Pavia University (PU), Salinas (SAs), and WHU-Hi-LongKou (LK). The details of these datasets are summarized in Table I.

1) Indian Pines: The IP dataset was acquired in June 1992 by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor over an agricultural test site in northwestern Indiana, USA. The imagery consists of 145×145 pixels with a spatial resolution of approximately 20 m/pixel. It comprises 200 spectral bands that cover the wavelength range of 0.4–2.5 μm. The ground truth (GT) includes 16 mutually exclusive land-cover categories, primarily representing various agricultural crops.

2) Pavia University: The PU dataset was collected by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over Pavia, northern Italy. The scene has a spatial dimension of 610×340 pixels with a fine spatial resolution of 1.3 m/pixel. After excluding noisy bands, 103 spectral bands remain, spanning from 0.43 to 0.86 μm. The GT involves nine urban land-cover classes, such as trees, asphalt, and bricks.

3) Salinas: The SA scene was captured by the AVIRIS sensor over the Salinas Valley in California, USA, in 1998. It primarily covers an agricultural region with a spatial resolution of 3.7 m/pixel and an image size of 512×217 pixels. The dataset contains 204 spectral bands and includes 16-GT classes representing different vegetation types, including vegetables, vineyards, and bare soil.

4) WHU-Hi-LongKou: The LK dataset was acquired on July 17, 2018, in Longkou Town, Hubei Province, China, using a headwall nanohyperspec sensor mounted on a DJI Matrice 600 Pro autonomous aerial vehicle (AAV) platform. The imagery has a spatial size of 550×400 pixels with 270 spectral bands ranging from 0.4 to 1.0 μm. The GT covers nine distinct agricultural land-cover categories, providing a high-resolution benchmark for fine-grained classification.

B. Experimental Setups

1) Implementation Details: To ensure a fair and rigorous comparison, all competitive methods, including the proposed DDFGFormer, are optimized using the Adam optimizer and an initial learning rate of 5×10^{-4}. The total training process consists of 200 epochs. All models are implemented within the PyTorch 2.2.2 framework with CUDA 12.1, and are executed

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TABLE II

QUANTITATIVE RESULTS (ACC%±STD%) OF IP. BEST IN BOLD AND SECOND BEST WITH UNDERLINE

Class	CNN-based		Transformer-based				Frequency-based			
	HPDM-SPRN	NPADCN	SSFTT	GAHT	GSC-ViT	MASSFormer	FS-CGNet	SFDCT	ADCF-Net	DDFGFormer
1	**96.34 ± 4.53**	68.57 ± 6.26	66.10 ± 2.39	56.59 ± 11.86	72.20 ± 10.09	88.71 ± 6.42	71.19 ± 5.50	59.65 ± 12.11	85.37 ± 11.34	89.74 ± 9.25
2	97.61 ± 1.13	96.72 ± 1.80	90.97 ± 2.52	93.15 ± 2.51	92.91 ± 3.19	95.96 ± 1.02	94.13 ± 2.78	97.34 ± 1.01	93.30 ± 1.49	**98.47 ± 1.39**
3	96.49 ± 2.03	97.38 ± 0.79	91.50 ± 5.95	95.14 ± 2.34	96.99 ± 2.38	95.70 ± 1.94	95.70 ± 1.79	97.23 ± 1.20	93.61 ± 3.41	**98.67 ± 1.26**
4	95.40 ± 4.07	87.50 ± 4.43	91.17 ± 2.90	92.96 ± 4.76	90.55 ± 5.58	93.82 ± 0.41	85.28 ± 7.83	91.67 ± 7.55	80.42 ± 8.05	**98.71 ± 1.61**
5	**97.91 ± 1.82**	85.73 ± 9.87	93.10 ± 2.51	94.78 ± 2.58	95.08 ± 3.24	91.66 ± 4.79	95.84 ± 2.95	95.25 ± 1.62	96.16 ± 2.44	97.52 ± 0.96
6	98.65 ± 0.94	98.80 ± 0.65	98.08 ± 1.70	98.39 ± 1.19	98.57 ± 1.17	92.29 ± 2.00	**99.71 ± 0.24**	99.06 ± 0.14	99.67 ± 0.42	99.05 ± 0.68
7	82.80 ± 17.71	**97.60 ± 3.20**	89.60 ± 16.02	79.20 ± 17.60	89.60 ± 9.67	69.51 ± 5.33	86.33 ± 13.05	96.67 ± 4.71	85.20 ± 12.14	95.83 ± 7.45
8	99.86 ± 0.42	99.95 ± 0.09	99.84 ± 0.42	99.79 ± 0.42	99.51 ± 0.85	99.96 ± 0.02	**100.00 ± 0.00**	99.40 ± 0.85	**100.00 ± 0.00**	**100.00 ± 0.00**
9	65.56 ± 20.31	82.22 ± 17.71	53.89 ± 15.53	37.22 ± 18.60	66.67 ± 21.37	62.49 ± 2.11	86.56 ± 13.43	78.55 ± 6.69	83.33 ± 19.40	**93.53 ± 8.09**
10	96.97 ± 1.49	94.40 ± 3.92	86.70 ± 4.07	89.82 ± 5.03	91.36 ± 4.85	95.10 ± 0.56	93.43 ± 3.46	94.33 ± 0.76	95.27 ± 1.45	**98.01 ± 1.33**
11	97.58 ± 0.63	98.02 ± 1.19	94.77 ± 1.78	96.93 ± 1.50	97.29 ± 1.00	97.86 ± 0.73	97.97 ± 0.68	98.59 ± 0.86	96.93 ± 1.27	**99.10 ± 0.84**
12	96.95 ± 2.14	**97.63 ± 1.07**	88.16 ± 6.11	92.87 ± 2.97	92.43 ± 2.37	89.21 ± 4.74	95.37 ± 3.13	92.65 ± 2.06	83.54 ± 5.14	97.16 ± 1.79
13	97.68 ± 3.01	97.65 ± 2.68	99.51 ± 0.51	98.49 ± 2.09	**99.78 ± 0.26**	98.84 ± 0.39	98.66 ± 1.67	98.99 ± 1.03	99.15 ± 0.62	99.43 ± 0.68
14	**99.87 ± 0.18**	99.36 ± 0.49	97.26 ± 1.30	97.63 ± 1.87	97.81 ± 1.58	97.41 ± 1.25	99.41 ± 0.38	98.92 ± 0.64	97.81 ± 2.11	99.69 ± 0.45
15	**99.02 ± 1.64**	87.07 ± 5.26	87.44 ± 6.22	93.11 ± 5.68	90.52 ± 6.51	94.50 ± 6.39	90.22 ± 5.59	94.83 ± 0.45	89.63 ± 5.37	98.26 ± 2.30
16	81.55 ± 7.27	95.76 ± 5.79	91.43 ± 10.51	92.62 ± 4.48	92.86 ± 8.23	**97.85 ± 0.09**	88.94 ± 8.05	92.97 ± 2.35	97.50 ± 3.03	93.29 ± 6.10
OA(%)	96.83 ± 0.27	96.30 ± 0.94	93.01 ± 0.83	94.91 ± 0.82	95.36 ± 0.59	96.20 ± 0.28	96.14 ± 0.71	96.99 ± 0.27	94.92 ± 0.93	**98.64 ± 0.31**
AA(%)	93.76 ± 1.94	92.77 ± 2.65	88.72 ± 2.75	88.04 ± 1.44	91.51 ± 2.52	91.35 ± 0.91	92.42 ± 1.94	92.92 ± 2.37	92.33 ± 2.05	**97.28 ± 1.28**
BF1(%)	91.47 ± 0.74	88.64 ± 0.87	87.75 ± 1.09	88.83 ± 0.89	90.12 ± 1.15	90.92 ± 0.88	89.57 ± 1.65	87.78 ± 0.72	84.23 ± 0.83	**93.81 ± 0.79**
Kappa×100	97.30 ± 0.31	95.78 ± 1.08	92.03 ± 0.95	94.19 ± 0.94	94.71 ± 0.68	95.55 ± 0.32	95.60 ± 0.82	96.57 ± 0.31	94.21 ± 1.07	**98.46 ± 0.35**

TABLE III

QUANTITATIVE RESULTS (ACC%±STD%) OF PU. BEST IN BOLD AND SECOND BEST WITH UNDERLINE

Class	CNN-based		Transformer-based				Frequency-based			
	HPDM-SPRN	NPADCN	SSFTT	GAHT	GSC-ViT	MASSFormer	FS-CGNet	SFDCT	ADCF-Net	DDFGFormer
1	98.76 ± 1.24	96.18 ± 1.73	95.32 ± 0.91	96.16 ± 2.29	94.65 ± 2.71	98.00 ± 0.54	96.51 ± 2.42	97.06 ± 0.77	97.94 ± 0.21	**99.80 ± 0.16**
2	99.57 ± 0.31	98.89 ± 0.03	98.83 ± 0.02	98.99 ± 0.55	99.79 ± 0.21	99.04 ± 0.17	99.69 ± 0.22	98.85 ± 0.89	99.81 ± 0.04	**100.00 ± 0.00**
3	85.20 ± 8.93	82.32 ± 1.71	62.70 ± 5.60	82.23 ± 4.61	89.27 ± 3.95	83.72 ± 4.42	85.89 ± 3.06	80.60 ± 6.00	93.65 ± 1.16	**97.57 ± 1.14**
4	86.37 ± 1.73	92.37 ± 2.69	**96.77 ± 2.01**	95.05 ± 1.07	95.98 ± 2.24	89.65 ± 2.10	91.88 ± 1.76	93.88 ± 3.53	90.90 ± 0.19	96.18 ± 1.81
5	99.77 ± 0.08	99.55 ± 0.45	99.92 ± 0.04	**100.00 ± 0.00**	96.17 ± 0.70	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	99.77 ± 0.02	99.47 ± 0.50
6	**100.00 ± 0.00**	98.23 ± 0.77	94.58 ± 0.89	90.10 ± 1.64	94.94 ± 1.64	97.18 ± 1.36	89.61 ± 2.44	94.26 ± 2.94	96.99 ± 3.16	**100.00 ± 0.00**
7	93.01 ± 0.40	91.27 ± 0.70	85.27 ± 12.08	76.64 ± 13.16	94.56 ± 0.19	96.84 ± 1.26	95.03 ± 3.45	91.51 ± 1.27	96.79 ± 2.35	**98.76 ± 0.26**
8	99.12 ± 0.52	90.44 ± 7.61	96.79 ± 1.73	85.01 ± 6.06	95.94 ± 2.10	97.33 ± 3.14	81.80 ± 5.34	90.66 ± 3.39	93.83 ± 0.36	**99.41 ± 0.37**
9	86.61 ± 1.55	84.61 ± 0.59	**93.57 ± 0.49**	89.50 ± 0.28	86.13 ± 1.89	89.76 ± 3.47	86.41 ± 6.77	93.51 ± 4.71	90.91 ± 0.71	92.65 ± 0.81
OA(%)	97.54 ± 0.60	96.65 ± 0.57	95.32 ± 2.10	94.55 ± 0.35	96.97 ± 0.42	96.21 ± 0.56	94.80 ± 1.55	95.76 ± 0.98	97.01 ± 0.07	**99.17 ± 0.23**
AA(%)	94.27 ± 1.39	92.65 ± 0.86	91.53 ± 3.05	90.46 ± 1.82	94.16 ± 1.05	94.61 ± 0.95	91.87 ± 2.68	93.37 ± 1.78	95.59 ± 0.16	**98.21 ± 1.19**
BF1(%)	91.79 ± 0.59	91.18 ± 0.71	90.38 ± 1.31	89.53 ± 1.92	92.21 ± 0.81	91.93 ± 0.58	89.08 ± 1.87	86.34 ± 2.46	92.01 ± 1.75	**94.46 ± 0.51**
Kappa×100	96.74 ± 0.80	96.40 ± 0.76	95.29 ± 1.03	92.73 ± 0.50	95.98 ± 0.03	94.95 ± 0.75	93.06 ± 2.09	94.38 ± 1.31	96.78 ± 0.25	**98.90 ± 0.30**

on a workstation equipped with an Intel i5-12490F CPU and an NVIDIA GeForce RTX 4070Ti GPU.

2) Data Partitioning: The datasets are randomly partitioned into training, validation, and testing sets. For the IP dataset, we select 5% of the labeled samples for training, 5% for validation, and the remaining 90% for testing. For the SA, PU, and LK datasets, the training and validation ratios are set to 0.5%, with the remaining 99% used for evaluation. The specific number of samples for each category across all datasets is detailed in Table I. To mitigate the impact of random initialization, all experiments are repeated ten times independently, and the mean values with standard deviation are reported as the final result.

3) Evaluation Metrics: Four widely used quantitative indices are employed to evaluate the classification performance: overall accuracy (OA), average accuracy (AA), boundary $F1$-score (BF1), and the kappa coefficient (kappa). Furthermore, to provide a qualitative assessment, we present the full classification maps along with zoom-in views of key regions. These visualizations enable direct comparison of spatial consistency and boundary preservation.

4) Competing Methods: To ensure a comprehensive and fair comparison, we select nine representative HSI classification methods from three mainstream paradigms. CNN-based methods (HPDM-SPRN [5] and NPADCN [22]) are included to represent local spatial–spectral modeling, providing a reference for evaluating whether DDFGFormer can further improve boundary and detail preservation beyond strong local feature extractors. Transformer-based methods (SSFTT [25], GAHT [50], GSC-ViT [26], and MASSFormer [6]) are used to represent global dependence modeling, providing a comparison with representative self-attention-based frameworks. Frequency-domain methods (FS-CGNet [38], SFDCT [37], and ADCF-Net [33]) are selected as the most directly related competitors, providing a direct evaluation of the proposed dual-decoupled frequency modeling against existing frequency-enhancement strategies. All competing methods follow the hyperparameter settings recommended in their original papers.

C. Comparison With SOTA Methods

1) Quantitative Analysis: Tables II–V illustrate the comparison of DDFGFormer with nine representative SOTA methods on four benchmark datasets. The best and second-best results are marked in bold and underlined, respectively. Overall, DDFGFormer achieves the best OA, AA, BF1, and kappa on all datasets, indicating its effectiveness across different HSI scenes. On IP, DDFGFormer obtains 98.64% in terms of OA, outperforming the best competing method SFDCT by 1.65%. The improvement is particularly evident for several

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TABLE IV

QUANTITATIVE RESULTS (ACC%±STD%) OF SA. BEST IN BOLD AND SECOND BEST WITH UNDERLINE

Class	CNN-based		Transformer-based				Frequency-based			
	HPDM-SPRN	NPADCN	SSFTT	GAHT	GSC-ViT	MASSFormer	FS-CGNet	SFDCT	ADCF-Net	DDFGFormer
1	**100.00 ± 0.00**	99.65 ± 0.50	98.44 ± 0.21	96.85 ± 2.03	**100.00 ± 0.00**	**100.00 ± 0.00**	99.98 ± 0.02	99.92 ± 0.08	**100.00 ± 0.00**	**100.00 ± 0.00**
2	99.99 ± 0.01	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	99.72 ± 0.40	99.86 ± 0.14	**100.00 ± 0.00**	**100.00 ± 0.00**
3	**100.00 ± 0.00**	92.86 ± 5.14	75.41 ± 4.61	94.15 ± 3.33	99.74 ± 0.30	99.08 ± 0.58	98.70 ± 1.52	99.41 ± 0.23	99.54 ± 0.10	99.89 ± 0.50
4	98.44 ± 1.05	95.57 ± 0.33	**100.00 ± 0.00**	99.64 ± 0.37	99.78 ± 0.09	96.87 ± 1.03	99.23 ± 0.74	80.52 ± 5.99	99.28 ± 0.06	99.76 ± 0.18
5	99.51 ± 0.04	92.51 ± 4.41	95.21 ± 3.33	96.11 ± 1.98	80.23 ± 5.29	94.04 ± 0.65	81.63 ± 2.08	99.65 ± 0.26	97.25 ± 0.73	**99.86 ± 0.88**
6	95.96 ± 1.42	99.17 ± 0.04	98.31 ± 0.97	99.25 ± 0.49	99.37 ± 0.02	98.95 ± 0.72	99.35 ± 0.93	98.95 ± 0.05	98.72 ± 0.08	**99.89 ± 0.01**
7	99.97 ± 0.03	99.95 ± 0.04	98.96 ± 1.42	99.47 ± 0.35	99.92 ± 0.05	99.89 ± 0.12	99.65 ± 0.49	99.20 ± 0.43	98.20 ± 0.43	**100.00 ± 0.00**
8	91.81 ± 2.56	92.96 ± 1.42	80.73 ± 5.42	92.75 ± 1.97	91.91 ± 2.55	91.90 ± 1.69	87.97 ± 3.45	93.18 ± 1.68	99.27 ± 0.28	**99.42 ± 1.62**
9	**100.00 ± 0.00**	99.65 ± 0.49	99.53 ± 0.39	99.31 ± 0.97	**100.00 ± 0.00**	99.69 ± 1.40	**100.00 ± 0.00**	99.93 ± 0.02	**100.00 ± 0.00**	**100.00 ± 0.00**
10	96.86 ± 1.05	90.35 ± 9.48	94.14 ± 2.09	95.89 ± 2.27	95.69 ± 2.57	94.76 ± 0.25	91.40 ± 3.81	97.84 ± 0.25	97.04 ± 0.05	**98.23 ± 0.34**
11	96.12 ± 3.22	97.63 ± 0.96	85.53 ± 1.08	94.51 ± 4.73	96.50 ± 3.38	98.70 ± 0.12	89.12 ± 2.32	94.00 ± 3.36	97.45 ± 0.62	**100.00 ± 0.00**
12	**100.00 ± 0.00**	99.08 ± 0.02	98.95 ± 0.01	98.43 ± 2.22	98.69 ± 1.40	99.32 ± 0.07	99.39 ± 0.62	98.61 ± 0.86	99.63 ± 0.01	99.81 ± 0.10
13	99.56 ± 0.24	97.93 ± 0.73	98.75 ± 0.83	92.54 ± 1.55	99.78 ± 0.05	98.34 ± 5.04	98.31 ± 2.39	96.43 ± 2.90	81.37 ± 1.63	**99.93 ± 0.10**
14	99.06 ± 0.66	98.65 ± 0.81	98.87 ± 0.54	98.55 ± 0.96	**100.00 ± 0.00**	93.62 ± 1.19	99.12 ± 0.25	99.66 ± 0.34	99.43 ± 0.06	**100.00 ± 0.00**
15	89.67 ± 7.04	88.57 ± 4.39	90.49 ± 3.91	92.24 ± 2.91	83.07 ± 4.58	90.59 ± 0.98	84.67 ± 2.98	78.21 ± 1.71	87.41 ± 0.27	**97.83 ± 0.06**
16	98.99 ± 0.61	98.49 ± 1.39	90.67 ± 2.30	94.19 ± 4.17	87.26 ± 3.62	99.49 ± 0.03	89.01 ± 9.50	99.56 ± 0.36	99.16 ± 0.02	**99.98 ± 0.03**
OA(%)	96.51 ± 0.28	95.62 ± 1.15	95.10 ± 1.08	96.04 ± 0.92	96.24 ± 1.39	97.82 ± 0.18	93.20 ± 1.42	95.69 ± 2.33	97.39 ± 0.07	**99.07 ± 0.14**
AA(%)	97.87 ± 0.17	96.44 ± 0.95	94.02 ± 1.20	96.49 ± 1.20	95.75 ± 0.19	97.20 ± 0.23	94.83 ± 0.96	95.97 ± 1.06	97.11 ± 0.18	**99.46 ± 0.26**
BF1(%)	90.04 ± 0.65	88.57 ± 1.50	89.32 ± 1.95	91.89 ± 0.35	90.57 ± 0.97	91.78 ± 1.25	90.45 ± 0.88	90.98 ± 0.58	91.60 ± 1.28	**94.50 ± 0.75**
Kappa×100	96.42 ± 1.04	95.12 ± 1.28	95.00 ± 0.67	95.59 ± 1.02	93.58 ± 1.76	97.68 ± 0.20	92.43 ± 1.58	92.98 ± 2.58	97.09 ± 0.21	**98.96 ± 0.16**

TABLE V

QUANTITATIVE RESULTS (ACC%±STD%) OF LK. BEST IN BOLD AND SECOND BEST WITH UNDERLINE

Class	CNN-based		Transformer-based				Frequency-based			
	HPDM-SPRN	NPADCN	SSFTT	GAHT	GSC-ViT	MASSFormer	FS-CGNet	SFDCT	ADCF-Net	DDFGFormer
1	99.88 ± 0.04	99.95 ± 0.03	99.87 ± 0.01	99.89 ± 0.02	99.97 ± 0.01	97.08 ± 1.00	99.41 ± 0.17	99.79 ± 0.10	98.09 ± 0.01	**100.00 ± 0.00**
2	98.45 ± 1.04	97.38 ± 0.35	98.95 ± 0.21	**99.27 ± 0.18**	98.95 ± 0.53	98.51 ± 0.86	93.62 ± 6.04	95.93 ± 0.75	97.65 ± 0.05	99.18 ± 0.96
3	96.66 ± 3.69	94.81 ± 2.81	90.50 ± 3.03	94.81 ± 3.98	95.05 ± 3.65	89.22 ± 3.77	92.82 ± 0.72	94.22 ± 3.59	95.27 ± 0.21	**99.59 ± 3.95**
4	**99.78 ± 0.08**	99.74 ± 0.09	99.62 ± 0.08	99.65 ± 0.25	99.62 ± 0.24	94.85 ± 1.49	98.38 ± 0.35	98.50 ± 0.11	98.66 ± 0.02	99.76 ± 0.04
5	98.34 ± 0.33	92.23 ± 0.23	97.03 ± 0.82	88.16 ± 7.05	97.03 ± 0.92	98.28 ± 0.94	91.20 ± 9.22	93.77 ± 1.21	97.40 ± 0.16	**98.66 ± 0.76**
6	99.52 ± 0.07	99.09 ± 0.08	99.63 ± 0.04	99.77 ± 0.19	99.63 ± 0.54	99.03 ± 3.72	99.81 ± 0.14	99.38 ± 0.59	99.19 ± 0.02	**99.97 ± 0.01**
7	98.93 ± 0.02	96.93 ± 0.04	94.93 ± 0.02	95.88 ± 0.53	96.93 ± 0.04	96.61 ± 1.52	97.95 ± 0.02	97.96 ± 0.01	98.94 ± 0.04	**99.96 ± 0.01**
8	96.04 ± 0.65	94.27 ± 1.23	96.20 ± 1.20	96.46 ± 0.70	96.20 ± 1.82	95.47 ± 1.19	96.01 ± 2.05	95.41 ± 0.62	93.89 ± 2.35	**98.35 ± 3.55**
9	94.59 ± 1.46	95.45 ± 1.89	92.04 ± 2.93	92.14 ± 0.75	93.04 ± 4.23	95.26 ± 2.47	90.71 ± 0.09	86.05 ± 4.12	90.02 ± 0.07	**96.28 ± 1.28**
OA(%)	98.39 ± 0.45	97.07 ± 0.72	97.25 ± 0.64	97.15 ± 0.32	98.25 ± 0.36	96.35 ± 0.54	97.04 ± 0.65	97.19 ± 0.54	97.11 ± 0.24	**99.53 ± 0.27**
AA(%)	98.02 ± 0.35	96.65 ± 0.26	96.53 ± 1.29	96.23 ± 1.40	97.38 ± 0.21	96.03 ± 0.97	95.55 ± 2.04	95.67 ± 0.28	96.57 ± 0.02	**98.98 ± 0.81**
BF1(%)	89.88 ± 1.28	88.34 ± 0.86	86.81 ± 1.27	89.03 ± 0.98	90.66 ± 1.47	84.12 ± 2.94	87.71 ± 2.19	88.69 ± 0.60	89.54 ± 0.73	**92.91 ± 0.28**
Kappa×100	98.19 ± 0.08	96.85 ± 0.31	96.02 ± 0.38	96.88 ± 0.42	98.02 ± 0.39	95.37 ± 0.68	96.28 ± 0.85	96.93 ± 0.23	96.58 ± 0.21	**99.50 ± 0.07**

spectrally similar or sample-limited classes. For example, DDFGFormer achieves 98.01% and 99.10% accuracy on soybean-notill and -mintill, respectively, which are higher than the frequency-based method ADCF-Net by 2.74% and 2.17%. It also improves the Oats class to 93.53%, showing better robustness for categories with very limited training samples. On PU, DDFGFormer reaches 99.17% OA and shows obvious advantages on confusing urban classes. For instance, it achieves 97.57% on Gravel and 98.76% on Bitumen, outperforming most competing methods by a large margin. On SA, the proposed method achieves 99.07% OA and improves challenging agricultural classes such as Grapes-untrained and Vineyard-untrained, which usually suffer from spectral similarity and intraclass variability. On LK, DDFGFormer maintains stable performance with a low standard deviation, indicating its robustness in high-resolution agricultural scenes. Meanwhile, DDFGFormer achieves the highest BF1 scores on all four datasets, demonstrating its effective boundary preservation in diverse HSI scenes. These results suggest that the proposed amplitude-phase decoupling and frequency-guided refinement not only improve overall classification accuracy but also enhance discrimination for spectrally similar, boundary-sensitive, and sample-limited categories.

2) Qualitative Evaluation: To visually evaluate the classification performance, full-scene classification maps of the four datasets are shown in Figs. 4–7, with representative subregions enlarged for detailed comparison. Overall, DDFGFormer generates more spatially coherent maps, with clearer class boundaries than the competing methods. On IP, the enlarged soybean-mintill regions show more homogeneous predictions, suggesting improved discrimination for spectrally similar categories. On PU, DDFGFormer better preserves boundaries among urban land-cover types, such as roads, buildings, and bare soil, where misclassification often occurs due to fragmented spatial structures. On SA and LK, the proposed method maintains more continuous elongated crop regions, which is important for agricultural scenes with strip-shaped fields and high intraclass similarity. These visual improvements are consistent with the quantitative results. They indicate that frequency-guided spatial refinement can enhance regional consistency while preserving boundary-sensitive details.

D. Parameter Analysis

1) Influence of Patch Size: The spatial dimension of the input patches is a critical hyperparameter that balances contextual richness against pixelwise interference. To investigate its influence on classification performance, we varied the patch size from 7×7 to 19×19 while keeping other parameters constant. As illustrated in Fig. 8(a), the four datasets exhibit distinct evolutionary trends. For IP and PU datasets,

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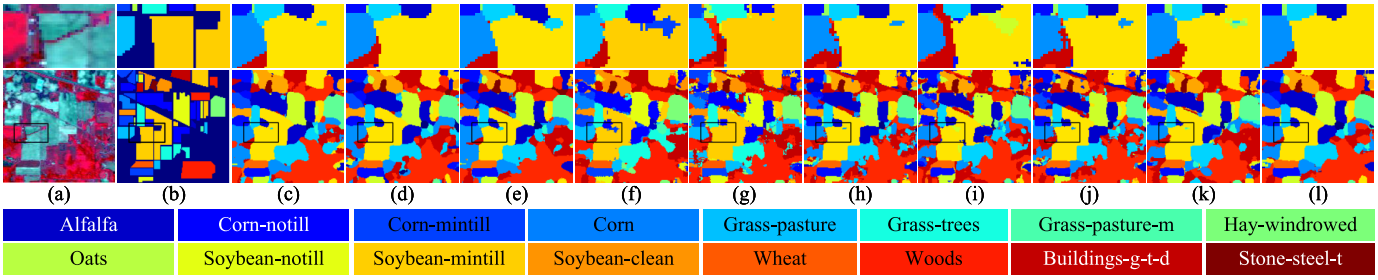


Fig. 4. Classification maps of different methods on the IP dataset. (a) False-color image. (b) GT. (c) HPDM-SPRN. (d) NPADCN. (e) SSFTT. (f) GAHT. (g) GSC-ViT. (h) MASSFormer. (i) FS-CGNet. (j) SFDCT. (k) ADCF-Net. (l) DDFGFormer.

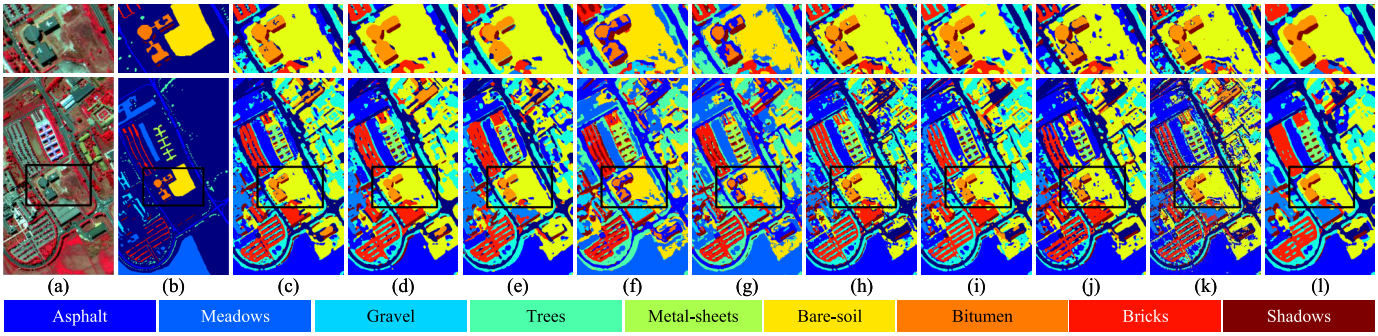


Fig. 5. Classification maps of different methods on the PU dataset. (a) False-color image. (b) GT. (c) HPDM-SPRN. (d) NPADCN. (e) SSFTT. (f) GAHT. (g) GSC-ViT. (h) MASSFormer. (i) FS-CGNet. (j) SFDCT. (k) ADCF-Net. (l) DDFGFormer.

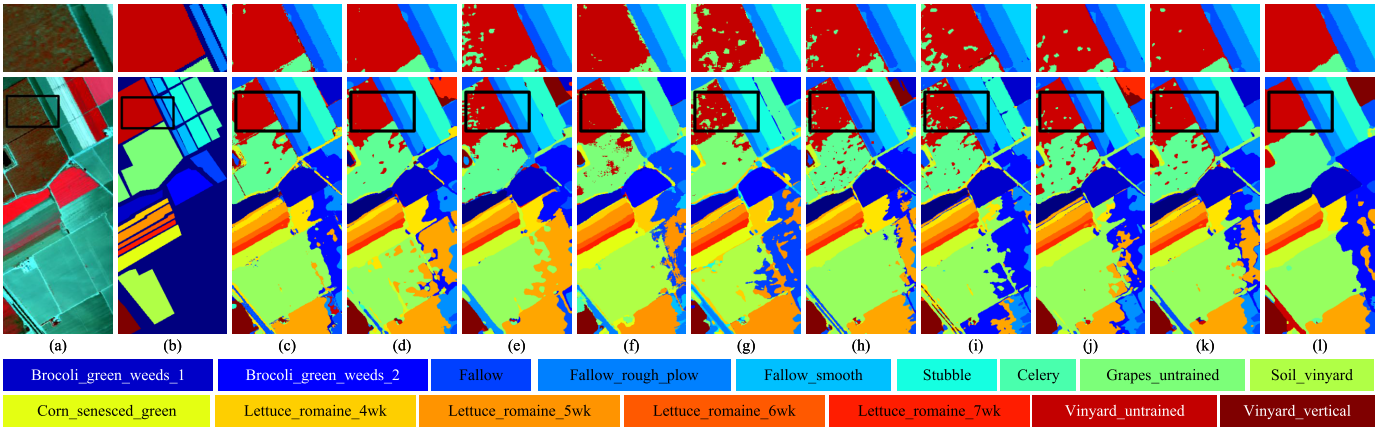


Fig. 6. Classification maps of different methods on the SA dataset. (a) False-color image. (b) GT. (c) HPDM-SPRN. (d) NPADCN. (e) SSFTT. (f) GAHT. (g) GSC-ViT. (h) MASSFormer. (i) FS-CGNet. (j) SFDCT. (k) ADCF-Net. (l) DDFGFormer.

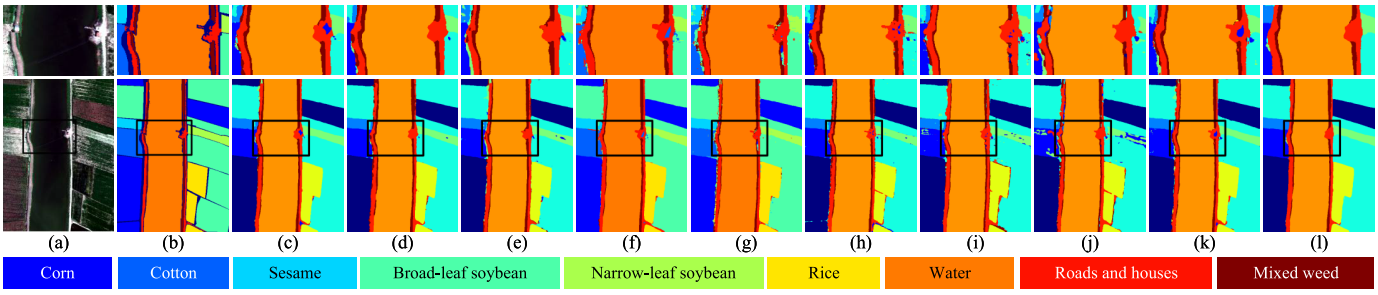


Fig. 7. Classification maps of different methods on the LK dataset. (a) False-color image. (b) GT. (c) HPDM-SPRN. (d) NPADCN. (e) SSFTT. (f) GAHT. (g) GSC-ViT. (h) MASSFormer. (i) FS-CGNet. (j) SFDCT. (k) ADCF-Net. (l) DDFGFormer.

the OA initially increases with larger patch sizes, suggesting that incorporating abundant spatial context facilitates the discrimination of complex land-cover types. However, a noticeable performance degradation occurs once the size

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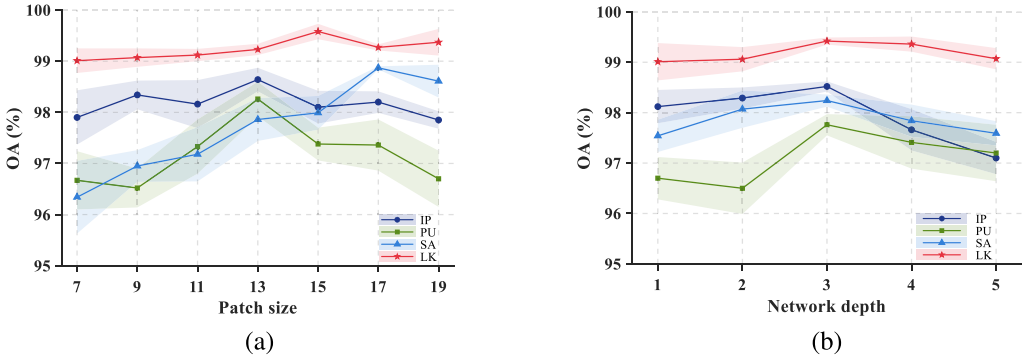


Fig. 8. Classification accuracies of the proposed DDFGFormer under different (a) patch sizes and (b) network depth, where the shaded region denotes the standard deviation.

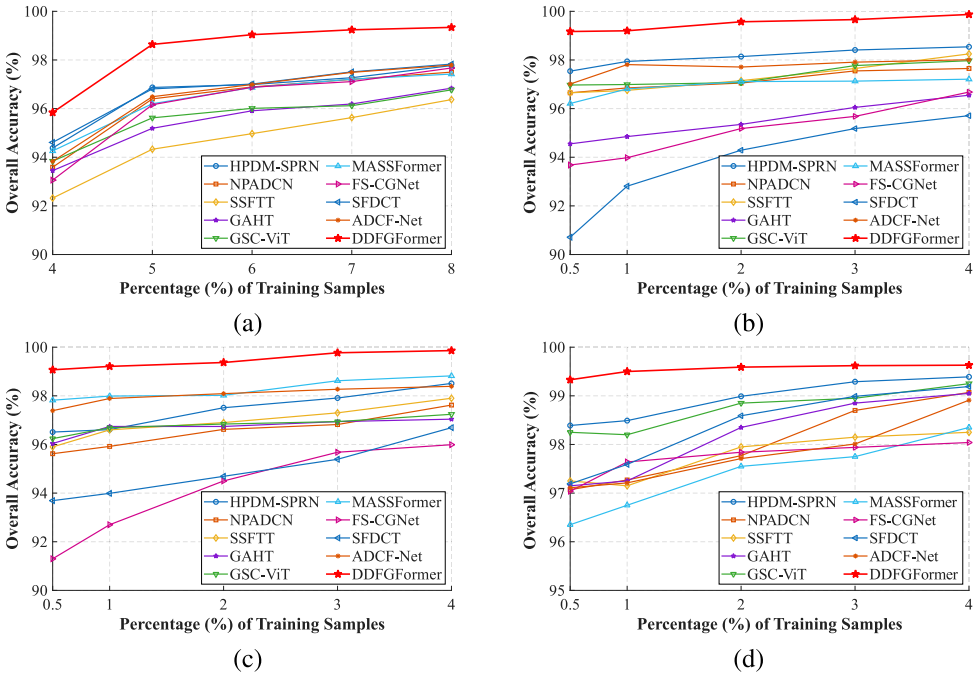


Fig. 9. OA of different methods with different percentages of training samples on (a) IP, (b) PU, (c) SA, and (d) LK.

exceeds the optimal threshold of 13 × 13. This phenomenon can be attributed to the inclusion of irrelevant interfering pixels and cross-category samples, which destroy the local spatial–spectral representations. The LK dataset follows a similar trend, reaching its optimal performance at 15×15. For the SA dataset, the OA continues to improve as the patch size increases. This observation arises from the presence of large-scale homogeneous land-cover distributions, where larger patches provide more stable long-range spatial dependencies without introducing excessive noise. Taking into account both classification accuracy and computational efficiency, patch sizes are empirically set to 13 for IP and PU, 15 for LK, and 17 for SA in subsequent experiments.

2) Influence of Network Depth: Network depth is an important hyperparameter of DDFGFormer. Although deeper networks can enhance feature abstraction, excessive stacking may lead to overfitting under limited training samples and degrade generalization. We vary the network depth from 1 to 5. As shown in Fig. 8(b), the OA on all four datasets first increases and then decreases, with the best performance achieved at depth 3. This suggests that a moderate depth is sufficient to capture effective spatial–frequency representations, whereas deeper configurations may introduce redundant features and overfitting, especially on IP and PU. On the contrary, SA and LK show relatively stable trends, possibly due to their larger homogeneous regions. Therefore, the network depth is set to 3 in all subsequent experiments to balance accuracy and computational cost.

E. Influence of Training Set Size

To evaluate the robustness of DDFGFormer under limited labeled samples, we conduct experiments with different training ratios on four benchmark datasets. In particular, the training ratio is varied from 4% to 8% on IP and from 0.5% to 4% on PU, SA, and LK. As shown in Fig. 9, DDFGFormer consistently achieves the highest OA across all sampling ratios. In the most challenging settings, such as 0.5% training samples on PU, SA, and LK, the proposed method still maintains stable performance, whereas several competing methods show

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TABLE VI
ABLATION ANALYSIS ON FOUR DATASETS. BEST IN BOLD

Case	Components					IP		PU		SA		LK	
	MSPM	Amp	Phase	APGP	FEFFN	OA(%)	BF1(%)	OA(%)	BF1(%)	OA(%)	BF1(%)	OA(%)	BF1(%)
1	✗	✗	✗	✗	✗	87.61±2.58	83.65±1.18	88.36±3.02	84.37±4.18	85.78±3.48	80.34±2.55	86.06±4.82	82.15±3.45
2	✓	✗	✗	✗	✗	89.25±2.62	85.37±1.55	91.12±3.86	86.18±3.06	88.15±5.27	82.89±4.23	87.65±3.79	83.98±2.26
3	✓	✓	✗	✗	✗	94.57±1.39	87.45±0.64	94.67±1.92	88.52±2.25	95.29±3.35	83.47±1.67	96.42±2.11	85.14±1.55
4	✓	✓	✓	✗	✗	96.56±1.49	89.25±1.37	96.08±1.78	89.13±1.49	97.56±1.31	86.63±1.94	98.32±0.82	87.83±1.36
5	✓	✓	✓	✓	✗	98.22±0.80	92.47±0.93	98.14±1.01	93.65±0.63	98.85±0.23	92.96±1.07	99.02±0.62	91.71±0.67
6	✓	✓	✓	✓	✓	**98.64±0.31**	**93.81±0.79**	**99.17±0.23**	**94.46±0.51**	**99.07±0.14**	**94.50±0.75**	**99.53±0.27**	**92.91±0.28**

accuracy degradation. As the training ratio increases, all methods generally improve, but DDFGFormer remains consistently superior and exhibits a smooth performance trend. This suggests that the proposed spatial–frequency decoupling and frequency-guided gating mechanisms help extract discriminative features from limited labeled samples, thereby improving robustness in small-sample HSI classification.

F. Ablation Study

1) Quantitative Ablation: To evaluate the contribution of each component, we design six incremental configurations, as reported in Table VI. Case 1 replaces the proposed MSPM with a conventional single-scale convolution operation, while all frequency-aware components are removed. This configuration serves as the basic spatial baseline. Case 2 further introduces MSPM based on Case 1 to evaluate the independent contribution of multiscale spatial feature modeling. Case 3 introduces amplitude modeling with high- and low- frequency decomposition to characterize frequency energy distributions at different scales. Case 4 further adds the phase branch and directly fuses amplitude and phase features. Case 5 introduces APGP on the basis of Case 4, aiming to verify whether amplitude-guided phase modulation is more effective than direct amplitude-phase fusion. Case 6 replaces the conventional FFN with the proposed FEFFN, forming the complete DDFGFormer.

The ablation results in Table VI reveal the progressive refinement of spatial and frequency representations introduced by different components. Compared with Case 1, Case 2 improves OA and BF1 by 1.64% and 1.72% on the IP dataset, respectively, while consistent improvements are also observed on the other three datasets. The BF1 gain substantially exceeds the OA gain, indicating that multiscale spatial priors contribute more to boundary localization than to global classification. By capturing local textures, neighboring structures, and boundary variations beyond single-scale convolution, MSPM reduces boundary ambiguity and improves spatial coherence. With amplitude modeling introduced in Case 3, further improvements in both OA and BF1 are obtained across all datasets, since amplitude information provides stable frequency energy cues that enhance the representation of local variations and global smooth structures. Case 4 further improves performance by incorporating phase information, as phase representations preserve structural positional information and facilitate more accurate boundary delineation and detail representation. However, the improvements over Case 3 are relatively modest, indicating that although phase carries complementary structural cues, directly fusing it without reliability estimation limits its contribution.

Compared with Case 4, Case 5 achieves OA and BF1 improvements of 1.66% and 3.22% on the IP dataset, respectively, with similar trends observed on the remaining datasets. The substantially larger BF1 gain than OA gain indicates that APGP primarily improves boundary preservation rather than global classification. By using high-frequency amplitude components as reliability priors, APGP stabilizes phase responses and suppresses unreliable structural information caused by noise and mixed pixels, leading to more reliable boundary features. Finally, Case 6 obtains the best performance with further improvements in OA and BF1 across all datasets, confirming the contribution of FEFFN. By continuously injecting discriminative frequency-domain information during feature propagation, FEFFN further refines fine-grained features while preserving complex spatial structures. Overall, these results provide quantitative evidence that MSPM, amplitude decomposition, phase modeling, APGP, and FEFFN contribute to spatial perception, frequency representation, phase reliability, and boundary preservation.

2) Boundary Error Visualization: To provide an intuitive analysis of boundary preservation, boundary error maps of six ablation cases on four datasets are presented in Fig. 10. As shown in the figure, a general reduction in boundary errors can be observed from Cases 1 to 6 across the four datasets. The early cases exhibit scattered false responses both within homogeneous regions and along object boundaries, while later cases generate cleaner and more continuous boundary structures. In particular, the complete DDFGFormer (Case 6) produces fewer residual errors and more accurate boundary delineation compared with other configurations. These visual observations are consistent with the BF1 improvements reported in Table VI, further validating the effectiveness of the proposed components in preserving spatial boundaries.

G. Complexity Analysis

Table VII reports the computational cost of different methods in terms of trainable parameters, FLOPs, training time, and testing time. DDFGFormer has a compact parameter scale, ranging from 159.68 to 258.70k across the four datasets. Taking the IP dataset as an example, DDFGFormer requires only 184.37k parameters and 2.60G FLOPs. Compared with FS-CGNet, SFDCT, and ADCF-Net, DDFGFormer reduces the parameter count by 63.00%, 65.03%, and 68.38%, respectively, and reduces FLOPs by 82.75%, 80.80%, and 77.23%, respectively while achieving higher classification accuracy.

The efficiency advantage mainly comes from the lightweight spatial–frequency refinement design. Existing frequency-domain methods usually introduce additional computation

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TABLE VII
COMPLEXITY ANALYSIS OF ALL METHODS ON FOUR DATASETS

Methods	IP				PU				SA				LK			
	Param(k)	FLOPs(G)	Train(s)	Test(s)	Param(k)	FLOPs(G)	Train(s)	Test(s)	Param(k)	FLOPs(G)	Train(s)	Test(s)	Param(k)	FLOPs(G)	Train(s)	Test(s)
HPDM-SPRN	204.49	4.34	75.96	22.50	185.52	38.79	59.08	34.15	205.39	22.96	89.92	51.28	293.20	24.52	87.15	31.51
NPADCN	364.42	5.35	141.06	3.01	217.65	25.54	39.09	26.55	370.93	27.45	78.43	21.69	365.27	30.43	92.43	28.34
SSFTT	172.50	4.32	42.39	5.40	175.20	32.45	38.15	12.20	173.58	13.42	55.61	8.90	215.40	25.41	62.42	14.10
GAHT	421.15	8.20	95.40	23.35	428.06	21.15	82.20	25.69	424.34	20.98	110.56	18.03	512.98	10.64	135.20	32.02
GSC-ViT	563.66	8.63	66.14	12.51	543.94	54.81	81.95	23.59	578.14	68.45	74.92	11.24	578.36	92.36	78.49	27.84
MASSFormer	246.12	9.16	46.91	2.44	304.24	14.89	22.78	5.38	345.21	17.59	31.54	7.64	360.55	25.67	97.16	18.49
FS-CGNet	498.35	15.07	82.01	34.65	409.01	56.74	73.66	34.64	608.52	63.88	113.06	67.07	611.75	79.64	139.46	62.39
SFDCT	527.22	13.54	65.59	23.56	508.55	48.76	98.67	43.56	587.99	52.54	120.90	13.01	591.05	85.45	146.08	27.03
ADCF-Net	583.06	11.42	78.84	38.90	523.43	63.74	85.54	41.22	588.34	61.78	147.17	72.28	596.72	80.86	152.28	74.40
DDFGFormer	184.37	2.60	23.31	19.50	159.68	27.63	47.80	21.59	195.82	20.58	74.26	36.06	258.70	24.21	67.88	32.31

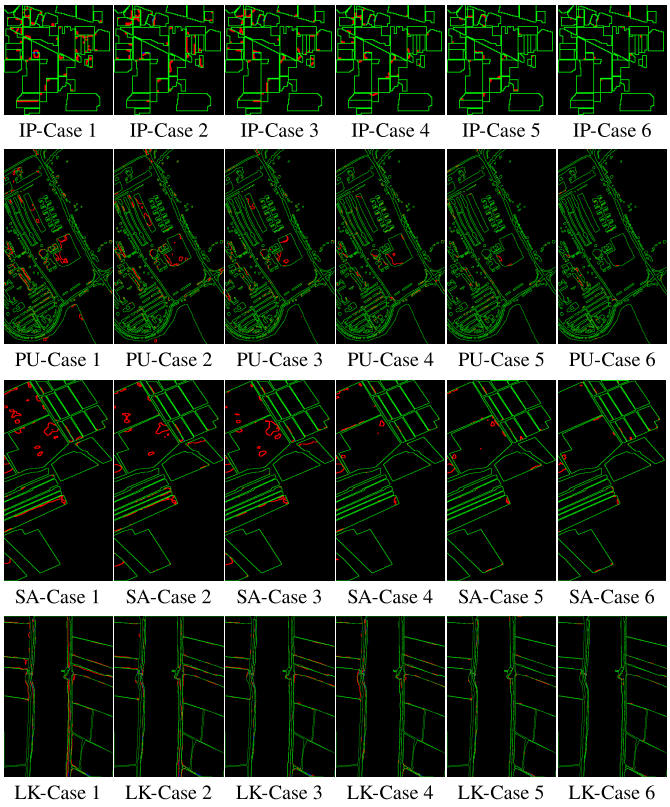


Fig. 10. Boundary error maps of different ablation cases on the four datasets. Green indicates perfectly matched boundaries between the prediction and the GT, red indicates extra noisy edges incorrectly predicted by the model, and blue indicates missed true boundaries that were not detected.

through dedicated frequency branches or heavy fusion modules. For example, SFDCT employs dual-branch transformer encoders with cross-attention fusion, FS-CGNet combines frequency extraction, channel attention, and cross-scale aggregation, and ADCF-Net uses separate real-valued and complex-valued streams. On the contrary, DDFGFormer embeds frequency analysis into a compact guidance mechanism. DDFA derives amplitude, phase, and high- and low-frequency components from FFT-based representations and mainly uses Gaussian filtering and elementwise modulation rather than heavy attention-based fusion. APGP further modulates phase with high-frequency amplitude priors through a compact gating mechanism. Although some lightweight CNN- or transformer-based methods may have lower testing time on certain datasets, DDFGFormer achieves a better balance among classification accuracy, boundary preservation, and computational efficiency.

V. CONCLUSION

In this article, we proposed DDFGFormer for HSI classification. By decoupling Fourier representations into amplitude and phase, further decomposing amplitude into high- and low-frequency components, and using high-frequency amplitude priors to adaptively guide phase modulation, the proposed DDFGFormer jointly exploits local details, global contours, and reliable phase-aware structural cues. Extensive experiments on four public HSI datasets demonstrate that DDFGFormer consistently outperforms state-of-the-art methods in classification accuracy and boundary preservation.

Despite its effectiveness, DDFGFormer still has several limitations that deserve further investigation. The frequency decomposition relies on fixed Gaussian filters whose bandwidth may not adapt to datasets with diverse spatial resolutions and scene complexities. In addition, while APGP improves the stability of phase representations in common HSI scenes, its robustness under more extreme noise scenarios remains to be thoroughly validated. Future work will focus on adaptive frequency decomposition with learnable filter parameterizations and more robust phase guidance mechanisms to further enhance the practicality of frequency-guided HSI classification.

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